

Homework 2

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Problem 1. *Prove*

$$\int_0^\infty \sin(x^2) dx = \int_0^\infty \cos(x^2) dx = \frac{\sqrt{2}}{4}$$

Proof. Use a quarter circle contour.

$$\lim_{R \rightarrow \infty} \int_\gamma e^{-z^2} dz = \lim_{R \rightarrow \infty} \left[\int_0^R e^{-x^2} dx + \int_0^{\frac{\pi}{4}} Rie^{i\theta} e^{-R^2 e^{i2\theta}} d\theta - \int_0^R e^{i\frac{\pi}{4}} e^{-r^2 e^{i\frac{2\pi}{4}}} dr \right].$$

By Cauchy's theorem, because e^{-z^2} is holomorphic on the open set inside the contour, $\lim_{R \rightarrow \infty} \int_\gamma e^{-z^2} dz = 0$. So

$$\lim_{R \rightarrow \infty} \int_0^R e^{i\frac{\pi}{4}} e^{-r^2 e^{i\frac{2\pi}{4}}} dr = \lim_{R \rightarrow \infty} \left[\int_0^R e^{-x^2} dx + \int_0^{\frac{\pi}{4}} Rie^{i\theta} e^{-R^2 e^{i2\theta}} d\theta \right]$$

$$\lim_{R \rightarrow \infty} \int_0^R e^{-x^2} dx = \frac{1}{2} \lim_{R \rightarrow \infty} \int_{-R}^R e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.$$

By the integral inequality,

$$\begin{aligned} \left| \int_0^{\frac{\pi}{4}} Rie^{i\theta} e^{-R^2 e^{i2\theta}} d\theta \right| &\leq \int_0^{\frac{\pi}{4}} \left| Rie^{i\theta} e^{-R^2 e^{i2\theta}} \right| d\theta \\ &= \int_0^{\frac{\pi}{4}} \left| Rie^{i\theta} e^{-R^2 (\cos(2\theta) + i \sin(2\theta))} \right| d\theta \\ &= \int_0^{\frac{\pi}{4}} |i| |e^{i\theta}| |R| \left| e^{-R^2 (\cos(2\theta) + i \sin(2\theta))} \right| d\theta \\ &= \int_0^{\frac{\pi}{4}} |R| \left| e^{-R^2 \cos(2\theta)} \right| \left| e^{-R^2 i \sin(2\theta)} \right| d\theta \end{aligned}$$

$\left| e^{-R^2 i \sin(2\theta)} \right| = 1$ so, $\int_0^{\frac{\pi}{4}} |R| \left| e^{-R^2 \cos(2\theta)} \right| \left| e^{-R^2 i \sin(2\theta)} \right| d\theta = \int_0^{\frac{\pi}{4}} |R e^{-R^2 \cos(2\theta)}| d\theta$. As $R \rightarrow \infty$, the exponential term dominates and $|R e^{-R^2 \cos(2\theta)}| \rightarrow 0$.

Thus, $\lim_{R \rightarrow \infty} \int_0^{\frac{\pi}{4}} Rie^{i\theta} e^{-R^2 e^{i2\theta}} d\theta = 0$.

Thus,

$$\begin{aligned}
\frac{\sqrt{\pi}}{2} &= \lim_{R \rightarrow \infty} \int_0^R e^{i\frac{\pi}{4}} e^{-r^2} e^{i\frac{2\pi}{4}} dr \\
&= e^{i\frac{\pi}{4}} \lim_{R \rightarrow \infty} \int_0^R e^{-ir^2} dr \\
\frac{\sqrt{\pi}}{2} e^{-i\frac{\pi}{4}} &= \lim_{R \rightarrow \infty} \int_0^R e^{-ir^2} dr \\
&= \lim_{R \rightarrow \infty} \left[\int_0^R \cos(r^2) dr - i \int_0^R \sin(r^2) dr \right] \\
&= \int_0^\infty \cos(r^2) dr - i \int_0^\infty \sin(r^2) dr.
\end{aligned}$$

Thus

$$\int_0^\infty \cos(r^2) dr = \Re \left(\frac{\sqrt{\pi}}{2} e^{-i\frac{\pi}{4}} \right) = \Re \left(\frac{\sqrt{\pi}}{2} \left(\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}} \right) \right) = \frac{\sqrt{2\pi}}{4}$$

and

$$\int_0^\infty \sin(r^2) dr = \Im \left(\frac{\sqrt{\pi}}{2} e^{-i\frac{\pi}{4}} \right) = \Im \left(\frac{\sqrt{\pi}}{2} \left(\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}} \right) \right) = \frac{\sqrt{2\pi}}{4}$$

□

Problem 2.

Proof. Use a indented semicircle contour and integrate the function $f(x) = \frac{e^{ix}}{x}$

By Cauchy's Theorem,

$$\int_\gamma \frac{e^{iz}}{z} dz = \int_0^\pi Rie^{i\theta} \frac{e^{iRe^{i\theta}}}{Re^{i\theta}} d\theta + \int_{-R}^{-\epsilon} \frac{e^{ix}}{x} dx + \int_\epsilon^R \frac{e^{ix}}{x} dx + \int_\pi^0 \frac{\epsilon e^{i\theta}}{\epsilon e^{i\theta}} e^{i\epsilon e^{i\theta}} d\theta = 0$$

$$\int_0^\pi Rie^{i\theta} \frac{e^{iRe^{i\theta}}}{Re^{i\theta}} d\theta = \int_0^\pi ie^{iRe^{i\theta}} d\theta$$

By the Integral inequality,

$$\begin{aligned}
\left| \int_0^\pi ie^{iRe^{i\theta}} d\theta \right| &\leq \int_0^\pi |ie^{iRe^{i\theta}}| d\theta \\
&= \int_0^\pi |e^{iRe^{i\theta}}| d\theta \\
&= \int_0^\pi |e^{-R \sin(\theta)}| |e^{iR \cos(\theta)}| d\theta
\end{aligned}$$

$$|e^{iR \cos(\theta)}| = 1.$$

Thus

$$\left| \int_0^\pi i e^{iRe^{i\theta}} d\theta \right| \leq \int_0^\pi |e^{-R \sin(\theta)}| d\theta = 0$$

because as $R \rightarrow \infty$, the exponential term decays to 0. Thus $\int_0^\pi |e^{-R \sin(\theta)}| d\theta = 0$ and $\left| \int_0^\pi i e^{iRe^{i\theta}} d\theta \right| = 0$.

$$\int_\pi^0 \frac{\epsilon i e^{i\theta}}{\epsilon e^{i\theta}} e^{i\epsilon e^{i\theta}} d\theta = \int_\pi^0 i e^{i\epsilon e^{i\theta}} d\theta$$

As $\epsilon \rightarrow 0$, $e^{i\epsilon e^{i\theta}} = \cos(\epsilon e^{i\theta}) + i \sin(\epsilon e^{i\theta}) \rightarrow 1$.

Thus

$$\int_\pi^0 \frac{\epsilon i e^{i\theta}}{\epsilon e^{i\theta}} e^{i\epsilon e^{i\theta}} d\theta = \int_\pi^0 i e^{i\epsilon e^{i\theta}} d\theta = -i\pi$$

Let $u = -x$, $du = -dx$, thus $\int_{-R}^{-\epsilon} \frac{e^{ix}}{x} dx = -\int_\epsilon^R \frac{e^{-iu}}{u} du$.

Slight abuse of notation, but set u to x .

Thus

$$\int_{-R}^{-\epsilon} \frac{e^{ix}}{x} dx + \int_\epsilon^R \frac{e^{ix}}{x} dx = \int_\epsilon^R \frac{e^{ix}}{x} dx - \int_\epsilon^R \frac{e^{-ix}}{x} dx = \int_\epsilon^R \frac{e^{ix} - e^{-ix}}{x} dx = 2i \int_\epsilon^R \frac{\sin(x)}{x} dx$$

Thus $2i \int_\epsilon^R \frac{\sin(x)}{x} dx - i\pi = 0$, thus as $R \rightarrow \infty$ and $\epsilon \rightarrow 0$,

$$\int_\epsilon^R \frac{\sin(x)}{x} dx = \int_0^\infty \frac{\sin(x)}{x} dx = \frac{\pi}{2}$$

□

Problem 3.

Proof. Suppose f is holomorphic on the open set ω containing the unit disk except at a pole z_0 on the unit circle and $\forall x \in \mathbb{D}$, $f(x) = \sum_{n=0}^\infty a_n x^n$. By assumption, f converges on the unit disk meaning $\lim_{n \rightarrow \infty} |a_n| = 1$. f is meromorphic on the domain ω with a pole at z_0 , thus let m be multiplicity of the pole at z_0 and $g(z)$ is holomorphic on ω , so

$$f(z) = \sum_{n=-m}^0 \frac{b_n}{(z-z_0)^n} + g(z)$$

By Lemma 5.10 in Chapter 2 of Stein, $\frac{1}{(z-z_0)^m} = \sum_{n=1}^\infty \frac{z^n}{z_0^n}$. Let $c_n = \frac{1}{z_0^n}$.

$$\frac{d^m}{dz^m} \left[\frac{1}{(z-z_0)^m} \right] = \frac{(-1)^m m!}{(z-z_0)^m} = \sum_{n=m}^\infty c_n \frac{n!}{(n-m)!} z^{n-m}$$

$$\lim_{n \rightarrow \infty} \frac{c_n}{c_{n+1}} = \lim_{n \rightarrow \infty} \frac{\frac{1}{z_0^n}}{\frac{1}{z_0^{n+1}}} = z_0$$

$$\lim_{n \rightarrow \infty} \frac{c_n \frac{n!}{(n-m)!}}{c_{n+1} \frac{(n+1)!}{(n+1-m)!}} = \lim_{n \rightarrow \infty} \frac{\frac{n+1-m}{z_0^n}}{\frac{n+1}{z_0^{n+1}}} = z_0$$

$$\sum_{n=-m}^0 \frac{b_m}{(z-z_0)^m} = b_1 \sum_{n=1}^{\infty} c_n z^n + \dots + b_m \sum_{n=m}^{\infty} c_n \frac{n!}{(n-m)!} z^{n-m} = \sum_{n=1}^{\infty} d_n z^n.$$

Thus d_n is a linear combination of c_n . Thus $\lim_{n \rightarrow \infty} d_n/d_{n+1} = z_0$.

Since $g(z)$ is holomorphic on ω , $g(z) = \sum_{n=0}^{\infty} k_n z^n$. However since g is holomorphic, $\lim_{n \rightarrow \infty} k_n = 0$ since g converges.

On the $\mathbb{D}$,

$$f(z) = \sum_{n=0}^{\infty} a_n z^n = \sum_{n=0}^{\infty} (c_n + k_n) z^n$$

Thus $a_n = c_n + k_n$ and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n$.

Thus

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1}} = \lim_{n \rightarrow \infty} \frac{c_n}{c_{n+1}} = z_0$$

□

Problem 4.

Proof. Let $F : \mathbb{C} \rightarrow \mathbb{C}$ where, $\forall z \in \mathbb{C}$,

$$F(z) = \begin{cases} f(z) & \text{if } |z| < 1 \\ 1/\overline{f(\frac{1}{\bar{z}})} & \text{if } |z| \geq 1 \end{cases}.$$

For $z \in \mathbb{C} \ni |z| \geq 1$, $|\frac{1}{\bar{z}}| \leq 1$. Since we know, $f(z)$ is non vanishing on $\bar{D}$, we know $f(\frac{1}{\bar{z}}) \neq 0$. Thus F is bounded below by 0 $\forall z \in \bar{\mathbb{D}}$ and $\overline{f(\frac{1}{\bar{z}})} \neq 0$. Thus, $\frac{1}{\overline{f(\frac{1}{\bar{z}})}}$ is holomorphic. Thus since F is holomorphic $\forall z \in \mathbb{C} \ni |z| < 1$, by assumption, and F is holomorphic $\forall z \in \mathbb{C} \ni |z| \geq 1$. Thus F is holomorphic in $\mathbb{C}$.

Furthermore, $\forall z \in \mathbb{C} \ni |z| = 1$, $\overline{(\frac{1}{f(\frac{1}{\bar{z}})})} = \overline{(\frac{1}{f(z)})}$, since $\frac{1}{z} = \bar{z}$. By assumption, $|f(z)| = 1$. Thus, $\frac{1}{\overline{f(z)}} = \overline{f(z)}$, thus $\overline{(\frac{1}{f(\frac{1}{\bar{z}})})} = f(z)$. Thus, F is well defined $\forall z \in \mathbb{C} \ni |z| = 1 \implies \forall z \in \mathbb{C}$, F is well defined.

By Theorem 2.1 in Stein Chapter 1, since f is continuous on $\bar{\mathbb{D}}$, $\exists M, N \in \mathbb{R} \ni \forall z \in \bar{\mathbb{D}}, N \leq f(z) \leq M$. Since f is nonvanishing on the domain, $0 < N$. Thus, $\forall z \in \mathbb{C} \setminus \bar{\mathbb{D}}, \left| \frac{1}{\bar{z}} \right| \leq 1$, thus by the above, $N \leq f(\frac{1}{\bar{z}}) \leq M$. Thus $|F(z)| = \left| \frac{1}{\overline{f(\frac{1}{\bar{z}})}} \right| \leq N$. Furthermore since $\frac{1}{\bar{z}} \in \bar{\mathbb{D}}$, and f is non vanishing on the domain, $0 < \frac{1}{\overline{f(\frac{1}{\bar{z}})}} = F(z)$. Thus $F(z)$ is bounded below by 0 $\forall z \in \mathbb{C} \setminus \bar{\mathbb{D}}$. Thus $F(z)$ is bounded below by 0, $\forall z \in \mathbb{C}$.

Thus $\forall z \in \mathbb{C}, F(z) \leq M$, thus $F(z)$ is bounded above by M and below by 0. Thus by Liouville's Theorem, F must be a constant function so f must be a constant function as well. □

Problem 5a.

Proof. $\forall x, r \in \mathbb{R} \exists p, q \in \mathbb{Z} \ni \frac{2\pi p}{2^q} \in N_r(x)$, by choosing a large enough q and a appropriate p , $|x - \frac{2\pi p}{2^q}| \leq \epsilon \forall \epsilon > 0$.

$\forall z \in \mathbb{D}, z = e^{i\theta}$ for $\theta \in \mathbb{R}$. By the above $\exists \frac{2\pi p}{2^q} \in N_\epsilon(\theta) \implies \exists r e^{i\frac{2\pi p}{2^q}} \in N_s(z)$. Let $z' = r e^{i\frac{2\pi p}{2^q}}$. $f(z') = \sum_{n=0}^{\infty} r^{2^n} e^{i2^{n-q}(2\pi p)} = \sum_{n=0}^{q-1} r^{2^n} e^{i2^{n-q}(2\pi p)} + \sum_{n=q}^{\infty} r^{2^n} e^{i2^{n-q}(2\pi p)}$. $\forall n \in \mathbb{N} \ni n \geq q$, $e^{i2^{n-q}(2\pi p)} = 1$. Thus $f(z') = \sum_{n=0}^{q-1} r^{2^n} e^{i2^{n-q}(2\pi p)} + \sum_{n=q}^{\infty} r^{2^n}$. $\lim_{r \rightarrow 1} \left[\sum_{n=q}^{\infty} r^{2^n} \right] = \infty = DNE$. Thus as $r \rightarrow 1$, $f(z') \rightarrow \infty$. Let g be a analytic function such that $g = f$ on $\mathbb{D} \cap N_\epsilon(\theta)$. As $r \rightarrow 1$, $g(z') \rightarrow \infty$. Thus by contradiction, g cannot be analytic since the power series of g is not locally convergent on z . Thus z is not a regular point. Since $\forall z \in \delta\mathbb{D}$, z is not regular, thus f cannot be analytically continued. $\square$

Problem 5b.

Proof. Let $z \in \delta D$. Let $\epsilon > 0$ be a real. There exists a large enough $N \ni \sum_{i=N+1}^{\infty} 2^{-i\alpha} \leq \frac{\epsilon}{2}$. Let $t \in D$, $|f(z) - f(t)| = \left| \sum_{i=0}^{\infty} 2^{-n\alpha} z^{2^n} - t^{2^n} \right| \leq \sum_{i=0}^N |z^{2^n} - t^{2^n}| + \frac{\epsilon}{2}$. Since z^{2^n} is continous $\forall n \in \mathbb{N}$, $\exists \delta \ni |z - t| \leq \delta$, such that $\sum_{i=0}^N |z^{2^n} - t^{2^n}| < \frac{\epsilon}{2}$. Thus for every $t \ni |z - t| \leq \delta$, $|f(z) - f(t)| < \epsilon$. Thus f is continous on δD , unit circle.

Assume the contrary f can be be analytically continued, thus $\exists w \in \delta D$ and a neighborhood U and a analytic function g such that $f = g$ on $D \cap U$.

Let $g_n(z) := z^n g^{(n)}(z)$. Suppose g_n is analytic. g_{n+1} is made by differentiating g_n and then multiplying z thus g_{n+1} is analytic. Thus by induction, $\forall n \in \mathbb{N}$, g_n is analytic. Since $g(z) = \sum_{i=0}^{\infty} 2^{-n\alpha} z^{2^n}$ on $D \cap U$, thus $g_m(z) = \sum_{n=0}^{\infty} 2^{n(m-\alpha)} z^{2^n}$ on $D \cap U$. Choose $N \in \mathbb{N} \ni N > \alpha$. $g_N(z) = \sum_{n=0}^{\infty} 2^{n(N-\alpha)} z^{2^n}$ for $z \in D \cap U$ is analytic.

However g_N is nowhere continous on δD . $\forall z \in \delta D$, use the trick from 5a, $\exists p, q \in \mathbb{Z} \ni e^{i\frac{2\pi p}{2^q}} \in V_s(z)$. $g_N(r e^{i\frac{2\pi p}{2^q}}) = \sum_{n=0}^{b-1} 2^{n(N-\alpha)} r^{2^n} e^{2\pi i 2^{b-n} p} + \sum_{n=b}^{\infty} 2^{n(N-\alpha)} r^{2^n} e^{2\pi i 2^{b-n} p}$ diverges as $r \rightarrow 1$. Thus $g_N(z)$ diverges. Thus g_N cannot be analytic and thus g cannot be analytic. Thus f cannot be analytically continued. $\square$